Webb Research Statement 2026

My *research agenda has developed through three interconnected streams* of scholarship examining how universities can continually better prepare engineers to address complex societal challenges by supporting their experiences of "critical tensions," or challenges necessary for learning.

The first strand of my research agenda **applies organizational systems theory to understand how academic ecosystems influence student and faculty researcher identity, motivation, and development**. In practical terms, this means examining how the structures universities use to organize themselves into systems—like departments, curriculum, funding mechanisms, tenure requirements, publishing expectations—actively shape whether and how graduate students, postdocs, and faculty develop and sustain certain scholarly identities and research agendas. My work in this space has evolved from initially analyzing individual interdisciplinary grad students' identity formation using Future Possible Selves theory to increasingly systems-oriented approaches employing Ecological Systems Theory (EST) and Latucca and Pollard's (2016) framework for influences on faculty decision-making for instructional change [1], [2], [3], [4]. This progression began with longitudinal studies revealing participation in interdisciplinary grad programs does not guarantee interdisciplinary identity development; rather, graduate students have to actively navigate complex tensions between disciplinary and interdisciplinary demands shaped by multiple, often misaligned, institutional systems in order to develop. Through comparative work across institutions and countries, I've demonstrated that challenges for interdisciplinary scholarship persist across contexts, though they manifest differently depending on relevant national priorities, institutional cultures, and disciplinary norms [5], [6]. My dissertation [7] synthesized five years of research to map 12 critical "microsystems"—environments like departments, research labs, and funding structures—and analyze how their core functions interact to create supports, barriers, and contradictions for interdisciplinary graduate student development. Notably, this work shows that sustainable change in interdisciplinary grad education requires more than adding interdisciplinary programs to existing disciplinary contexts without integration; it demands coordinated transformation of how multiple academic systems function at their core, because isolated interventions are absorbed and often undermined by entrenched disciplinary structures. My two consecutive ASEE Best Student Paper Awards [8], [9] recognized this systems-based approach to understanding why interdisciplinary graduate education remains challenging despite decades of investment and how microsystem interactions create compounding effects that must be addressed holistically. Currently, I'm extending this thinking with Activity Systems Theory and relational-agency informed approaches to postdoctoral professional development, examining how postdocs develop capacity for instructional change within similarly complex academic ecosystems. My team and I have been developing a framework for understanding graduate and postdocs' experiences of "critical tensions" – challenging learning situations that are necessary for professional development—and how emerging scholars can be supported through them [10]. This critical tensions framing and systems approaches are especially salient for understanding development in emerging fields, where scholars must navigate uncharted territory and push against entrenched systems while pioneering new forms of scholarship without established precedents to guide them.

The second research stream **focuses on advancing methodological approaches for studying complex educational and engineering systems.** First, traditional qualitative interview methods in both engineering and engineering education research, while producing rich insights, struggle to process the volume and complexity of data needed to understand systems-level educational phenomena across multiple contexts and timeframes. In response, I've helped develop generative AI-enabled methods for qualitative analysis that maintain analytical depth while enabling scale. My early work in this space helped establish methodological foundations in social media risk communication for how and when AI can augment, rather than replace, human expertise in qualitative analysis with GenAI [11]. Now, I'm extending that learning into collaborations on reflective practice assessment in undergraduate STEM education [12]. My current research group has been pioneering multi- and mixed-validation approaches that triangulate qualitative and quantitative methods of codebook validation to ensure both human and AI coders reliably identify theoretically grounded constructs in large-scale assessments of student reflection quality. We've found that without this rigor, automated assessment risks scaling error rather than insight. My dissertation also exemplifies this commitment to multi-theory research techniques for analyzing complex educational systems, additionally putting forth novel visualization techniques for longitudinal qualitative data analysis of student development and systems' influences on it. By integrating Ecological Systems Theory with Future Possible

Selves frameworks, I developed methods—including thematic trajectory mapping supported by temporal meta-matrix analysis of qualitative coding across multiple interviews as well as microsystem interaction diagrams—that make visible how academic systems work together to shape student development [13], [14].

My participatory action research (PAR) represents a complementary methodological approach furthering human-computer interaction research in not just engineering education, but also engineering design. Across projects with refugee and immigrant communities and hospital educators, I've co-developed educational technologies that center community expertise [15], [16], [17], [18], [19], [20], [21]. For example, we have developed a web application called "100 USCIS Civics Questions" that offers novel audio and text modes with multilingual support (previously existing applications were in English-only and did not offer audio options) and multiple study modes based on our collaborative design workshop findings. This PAR work contributes approaches for community-engaged design that move beyond user-centered design to genuinely co-construct technologies with the communities they serve, providing methodological foundations for transformation in engineering design education. PAR represents a critical counterbalance to traditional engineering pedagogies and research methods, ensuring that as we scale research through technology, we honor diverse forms of knowledge and expertise. I envision incorporating PAR approaches in my engineering classrooms to bridge the gap between research and practice that too often characterizes engineering education — creating courses where students don't just learn about community-engaged design but actually do it, contributing to real projects in their local areas. For example, on-campus infrastructures like electric power generators or local manufacturing processes can serve as project-based learning sites for students to design engineering improvements based on empirical data and community members' experience.

My third research stream **examines research-to-innovation cycles of discipline-based education research (DBER) in transforming engineering design education.** This work focuses on translating research insights into actionable change models for interdisciplinary programming, participatory design pedagogies, and active learning implementation. In practical terms, this means understanding not just what works in engineering education, but how change happens—and how to build capacity for sustained instructional innovation. My work on interdisciplinary graduate education informs this transformation agenda by revealing the complexity of instructional change in practice. Even well-funded, carefully designed interdisciplinary programs encounter resistance from entrenched disciplinary structures, and these systems insights also apply to implementing evidence-based teaching practices [7]. Additionally, our PAR book chapter examining my own and another student's development engineering students engaged in participatory action research provides evidence of how such experiences can transform students' identities and research [15]. However, my work on how to build DBER change capacity through postdoctoral professional development programs aimed at STEM higher education instructional change connects most directly [10]. Postdoctoral roles across DBER and STEM fields more broadly have been posed as drivers of change or innovation resulting from research, but little scholarship examines how postdocs develop this capacity for change. There is also scant research on postdoc-specific PD programs for this capacity development, but scholarship that does exist highlights both the promise of postdocs as innovation drivers and relational tensions that can constrain their work. These tensions usually stem from postdoc-faculty relationships, as postdocs navigate "misalignments" in expectations and have to seek necessary learning on-the-job in order to meet delegated aims. My research team and I are working to address these connected issues by first developing a conceptual framework for how graduate students and postdocs (emerging scholars) develop professional agency in relationships through experiences of "critical tensions." We are empirically testing the theory with interview data from DBER postdocs and faculty counterparts work together engaged in implementing DBER within STEM departments through structured teaching innovation grants [22]. The framework and expected findings can guide both postdoc-specific professional development program design and future empirical studies of how postdocs working with faculty —and by extension, their fields — develop capacity for evidence-based change.

Complementing this work on building change capacity is my budding research examining specific pedagogical innovations via the Scholarship of Teaching and Learning. My collaborative work on hybrid, hands-on engineering courses demonstrates how evidence-based teaching strategies can be adapted for remote and distributed learning contexts without sacrificing the experiential learning critical to engineering education, recognizing professional engineering programs in particular necessity online components. This research, focused on graduate level robotics, embedded systems, and IoT courses, contributes practical strategies for implementing active learning principles in constrained institutional contexts that can be readily adopted by faculty teaching similar courses [23]. We do this by recognizing transformation must account for the real systems in which teaching occurs. For example, our findings reveal how working professionals in distance learning cohorts exhibited deeper engagement than on-campus students when provided flexible, asynchronous access to hardware-intensive projects—demonstrating that effective pedagogical innovation requires understanding learner contexts and institutional constraints, not just importing evidence-based practices wholesale.

My research agenda is also deeply connected to my teaching and mentorship in STEM education. I bring the same systems thinking and critical tensions framework into my classroom and mentoring relationships — recognizing that students and emerging scholars learn best not when challenges are removed, but when they are scaffolded through them. My teaching draws directly on the evidence-based practices my research examines, allowing me to evaluate and refine those approaches from the inside.

Looking ahead, my research agenda will deepen and connect these three streams in several directions. First, I aim to develop a more precise theory of critical tensions as a mechanism for learning and professional development — one informed by Vygotsky's zone of proximal development. While my current work establishes that critical tensions are necessary for growth, pressing questions remain: What kinds of tensions are productive versus paralyzing? How much challenge is generative, and how much becomes an obstacle? And when in a scholar's development are particular tensions most likely to catalyze growth rather than derail it? Answering these questions requires longitudinal, comparative data across graduate students, postdocs, and faculty at multiple institutions — work I intend to pursue through expanded multi-institutional studies that can reveal how institutional context, disciplinary culture, and career stage interact to shape when and how tensions become learning opportunities. Second, I will continue advancing methodological tools for studying these phenomena at scale, including further development of GenAI-enabled qualitative methods and visualization techniques that make complex developmental trajectories legible across large, diverse datasets — enabling the kind of cross-institutional and cross-cultural comparisons that can move findings beyond single-context insights toward broader generalizability. Third, I am committed to closing the loop between my theoretical and empirical research through expanded Scholarship of Teaching and Learning work — designing, implementing, and rigorously evaluating pedagogical models that my systems and critical tensions research supports. For example, this means not only theorizing how interdisciplinary identity develops, but testing whether intentionally designed interdisciplinary course experiences or postdoctoral mentoring structures that surface and scaffold critical tensions actually produce the developmental outcomes the theory predicts. Together, these directions reflect a research agenda oriented toward both understanding how academic systems shape scholar development and actively building the evidence base for transforming them.

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